



Trapezoidal integration for the prediction step

- For a 1-d problem, we can write the prediction step as:

$$\begin{aligned}
 f(x_k | \mathbb{Z}_{k-1}) &= \int_{x_{\min}}^{x_{\max}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1} \\
 &= \int_{x_{\min}}^{x_{\max}} g(x_{k-1}; x_k, \mathbb{Z}_{k-1}) dx_{k-1} \\
 &\approx w \left[\frac{g(x_{\min}; x_k, \mathbb{Z}_{k-1}) + g(x_{\max}; x_k, \mathbb{Z}_{k-1})}{2} + \sum_{i=1}^{N_r-1} g(x_{\min} + i w; x_k, \mathbb{Z}_{k-1}) \right].
 \end{aligned}$$

- We are now interested in implementing this operation numerically.
 - We can't implement the integral over an infinite set of values of dx_{k-1} , so we must discretize the dummy variable of integration x_{k-1} over N_r regions (3rd line of eqn.).
 - Similarly, we cannot store all values of $f(x_k | \mathbb{Z}_{k-1})$ corresponding to an infinite set of x_k . Need to discretize $f(x_k | \mathbb{Z}_{k-1})$ by discretizing x_k into N_r regions.



Tradeoff in precision versus speed

- If we discretize x_k into N_r regions, then we must compute:

$$\begin{aligned}
 f(x_k | \mathbb{Z}_{k-1}) &\approx w \left[\frac{g(x_{\min}; x_k, \mathbb{Z}_{k-1}) + g(x_{\max}; x_k, \mathbb{Z}_{k-1})}{2} \right. \\
 &\quad \left. + \sum_{i=1}^{N_r-1} g(x_{\min} + i w; x_k, \mathbb{Z}_{k-1}) \right].
 \end{aligned}$$

for each of those N_r possible values of x_k .

- We have N_r additions for each of the N_r values of x_k : $\mathcal{O}(N_r^2)$ operations overall.
 - “Curse of dimensionality” when dealing with multi-dimensional problems: $\mathcal{O}(N_r^{2n})$.
- The limiting factor to computing an estimate of $f(x_k | \mathbb{Z}_k)$ is precision vs. speed:
 - We increase precision by increasing N_r ;
 - We increase speed by decreasing N_r .
- Brute-force integration not an appealing final solution, but gives high-quality results.



Tracking example introduction

- In the next lesson, we will look at Octave code to implement a 1-d Bayesian recursion.
- For now, we set up a sample problem to solve: Suppose that we wish to track time-varying x_k for a system modeled by:

$$\begin{aligned}
 x_{k+1} &= \alpha x_k + w_k \\
 z_{.k} &= 0.1 x_k^3 + v_k,
 \end{aligned}$$

where $w_k \sim \mathcal{N}(0, \sigma_w^2)$ and $v_k \sim \mathcal{N}(0, \sigma_v^2)$ are mutually uncorrelated and IID.

- When running a simulation, we need to select values for design parameters:
 - Number of simulation iterations: We will use $k = 0 \dots 200$.
 - Range of integration: We will use $x_{\min} = -3$ and $x_{\max} = 3$.
 - Number of regions for x_k : We will use $N_r = 500$ regions.
 - We will further define model parameters: $\sigma_w = 0.2$, $\sigma_v = 0.1$, and $\alpha = 0.99$.



Tracking example prediction step

- For the prediction step, numerically integrate to approximate:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}.$$

- The pdf $f(x_{k-1} | \mathbb{Z}_{k-1})$ is the output from the prior iteration: a vector of N_r values.
- We determine the pdf $f(x_k | x_{k-1})$ from the model state equation.
 - Since $x_k = \alpha x_{k-1} + w_{k-1}$, if x_{k-1} is considered to be known, x_k is the sum of a known constant and a Gaussian random variable $w_{k-1} \sim \mathcal{N}(0, \sigma_w^2)$.
 - Therefore, if x_{k-1} is known, x_k is Gaussian and its conditional pdf is determined by its mean and variance. $\mathbb{E}[x_k | x_{k-1}] = \alpha x_{k-1}$ and $\text{var}(x_k | x_{k-1}) = \sigma_w^2$.
- So, $f(x_k | x_{k-1}) \sim \mathcal{N}(\alpha x_{k-1}, \sigma_w^2)$. Can evaluate for any value of x_k via:

$$f(x_k | x_{k-1}) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left(-\frac{(x_k - \alpha x_{k-1})^2}{2\sigma_w^2}\right).$$



Tracking example update step

- For the time-update step, we directly compute:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- The pdf $f(x_k | \mathbb{Z}_{k-1})$ is the output of the prediction step, an N_r -element vector.
- We determine the pdf $f(z_k | x_k)$ from the model output equation.
 - Since $z_k = 0.1x_k^3 + v_k$, if x_k is considered to be known, z_k is the sum of a known constant and a Gaussian random variable $v_k \sim \mathcal{N}(0, \sigma_v^2)$.
 - Therefore, if x_k is known, z_k is Gaussian and its conditional pdf is determined by its mean and variance. $\mathbb{E}[z_k | x_k] = 0.1x_k^3$ and $\text{var}(z_k | x_k) = \sigma_v^2$.
- So, $f(z_k | x_k) \sim \mathcal{N}(0.1x_k^3, \sigma_v^2)$. Can evaluate for any value of z_k via:

$$f(z_k | x_k) = \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{(z_k - 0.1x_k^3)^2}{2\sigma_v^2}\right).$$



Tracking example additional details

- For the time-update step, note the denominator of:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- We can compute this conditional pdf via the law of total probability (a giant pain).
- Or, we can recognize that its only function is to ensure $f(x_k | \mathbb{Z}_k)$ integrates to 1.
- It is easiest to compute the numerator first, then divide that computation by its integral to normalize the pdf $f(x_k | \mathbb{Z}_k)$ to have an integral of one.
- Finally, note that the predict/update recursion requires that $f(x_0 | \mathbb{Z}_0)$ be specified, where:

$$f(x_0 | \mathbb{Z}_0) = \frac{f(z_0 | x_0) f(x_0 | \mathbb{Z}_{-1})}{f(z_0 | \mathbb{Z}_{-1})}.$$

- Since $\mathbb{Z}_{-1}$ is unknown, we assume that $f(x_0 | \mathbb{Z}_{-1}) = f(x_0) \sim \mathcal{N}(0, \sigma_w^2)$.



Summary

- In this lesson, you learned how to evaluate sub-components of the Bayesian recursion, which were abstract until this point.
- You have seen more details on how to compute the integral in the prediction step that estimates $f(x_k | \mathbb{Z}_{k-1})$.
 - This included an introduction to the “curse of dimensionality” that the particle filter will need to overcome eventually.
- You have seen how to derive and compute $f(x_k | x_{k-1})$ and $f(z_k | x_k)$.
- We are now ready to implement this example in Octave code to see the kinds of results for $f(x_k | \mathbb{Z}_k)$ we might expect.
 - Once we have $f(x_k | \mathbb{Z}_k)$, we can compute mean, median, mode, whatever.
 - Can plot these point estimates as functions of time, just like with a Kalman filter.